

Keerthi Sirapu

SOFTWARE ENGINEER

NY, USA | keerthi.b.sirapu@gmail.com | +1 (934) 263-1328 | [LinkedIn](#)

SUMMARY

Software Engineer with around 4 years of experience building scalable backend systems and cloud-native applications using Python (FastAPI, Flask, Django) across fintech, automotive, and sustainability domains. Skilled in designing microservices, event-driven architectures, and real-time data pipelines using Kafka, gRPC, and AWS services including Lambda, EC2, EKS, SQS, CloudWatch, and Step Functions. Adept at integrating backend services with modern front-end interfaces using React, Redux, and TypeScript, and improving data processing performance through optimized SQL and Elasticsearch queries. Experienced in containerized deployments with Docker and Kubernetes, implementing CI/CD pipelines with GitHub Actions, GitLab CI/CD, and Jenkins, and ensuring robust system reliability with Kibana, Jaeger, Dynatrace, and structured logging. Known for delivering clean, maintainable code, enhancing automation and observability, and collaborating in Agile teams to translate complex business requirements into efficient, production-ready solutions.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Node.js, SQL, Golang

Frameworks & Libraries: FastAPI, Flask, Django, React, React Hooks, Redux, Express.js, gRPC

Frontend Technologies: React.js, HTML5, CSS3

Backend & API Development: RESTful APIs, Microservices Architecture, Event-Driven Architecture

Cloud Platforms: AWS (Lambda, EC2, EKS, SQS, CloudWatch, Step Functions), Google Pub/Sub

Databases: PostgreSQL, Oracle, Redis, Elasticsearch

DevOps, CI/CD & Containers: Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Kafka

Testing & Automation: pytest, Jest, Cypress

Monitoring & Observability: Kibana, Jaeger, Dynatrace, ELF (Enterprise Logging Framework)

Version Control & Collaboration: Git, GitHub, GitLab, Jira, Agile/Scrum

PROFESSIONAL EXPERIENCE

BNY Financials | Software Engineer

NY, USA / Sep 2025 – Present

- Developed Python FastAPI services to process survey response streams and generate real-time financial insights for community banks, integrating a React-based analytics dashboard to visualize lending sentiment, liquidity trends, and regulatory readiness.
- Built Django admin modules and Redux-driven state management to configure survey scoring logic, regional segmentation rules, and institution-level benchmarking parameters used across the insights platform.
- Implemented Kafka pipelines to ingest market indicators, survey responses, and transactional metadata, while using AWS Lambda, CloudWatch, and Step Functions to automate data normalization, scoring workflows, and scheduled insight-generation tasks.
- Created Python utilities to integrate with Salesforce APIs for institution onboarding and outreach tracking and developed Node.js data transformation helpers to streamline bulk survey imports from partner banks.
- Optimized analytical queries across SQL and Elasticsearch, reducing time-to-insight for historical survey data, and improved UI reliability through Jest-based unit testing for interactive analytics components.
- Ensured platform stability using Docker, Kubernetes, Git, GitHub, and enterprise-grade observability with ELF and Kibana, strengthening compliance, auditability, and system resilience for financial institutions.

IUS Auto Inc | Software Engineer

Maricopa, AZ, USA / Aug 2024 – May 2025

- Built Python Flask and FastAPI microservices to support vehicle diagnostics, warranty validations, and service automation, integrating with a React front-end for dealership portals.
- Implemented gRPC-based Python services connected to Kafka topics for telematics streaming, and deployed ingestion workers on AWS EC2 and EKS (Kubernetes) clusters.
- Developed backend modules for VIN decoding, service history tracking, and dealership workflows, using TypeScript + Node.js to handle front-end data normalization.
- Designed scalable data pipelines using Python, Oracle, Elasticsearch, Redis, and optimized queries through JavaScript-based indexing scripts for faster retrieval.
- Containerized multiple diagnostics services using Docker, managed rollouts through Kubernetes and GitLab CI/CD for consistent multi-environment deployments.
- Automated ETL processes and model-prep workflows with Python scripts, and validated data quality using Cypress & pytest automation suites.
- Enhanced system observability with Kibana dashboards, Jaeger tracing, structured logging, improving debugging efficiency for dealership outages.

- Contributed Golang microservice optimized for high-throughput IoT event processing, ensuring compatibility with Python-based pipelines.

Accenture | Software Engineer

India | Jul 2021 – Aug 2023

- Developed Python FastAPI microservices for automated sustainability data ingestion and applied TypeScript-based validation utilities to improve accuracy across ESG indicators.
- Built internal Django + React (Hooks & Redux) dashboards for real-time emissions and compliance visibility, enhancing decision-making for sustainability teams.
- Designed Kafka + Google Pub/Sub event-driven pipelines to handle continuous ESG data flow, using AWS Lambda & SQS to orchestrate cloud-based transformations.
- Implemented secure REST APIs in Python, integrated with Node.js middleware for authentication and request routing between UI and backend.
- Optimized ESG data querying using Elasticsearch and PostgreSQL, while building JavaScript-based bulk import utilities for sustainability analysts.
- Automated CI/CD pipelines using GitHub Actions, Jenkins, Docker, and Kubernetes, ensuring reliable deployments across multi-region environments.
- Strengthened observability using Kibana, Jaeger, Dynatrace, with test coverage increased via PyTest + Jest for API and UI reliability.
- Collaborated in Agile sprints using Git/GitHub workflows, translating sustainability and compliance needs into scalable Python-driven architecture.

EDUCATION

Stony Brook University, Stony Brook, New York, USA
Master of Science in Data Science

Aug 2023 – May 2025

CVR College of Engineering, Telangana, India
Bachelor of Science in Computer Science

Jul 2017 – Jun 2021

PROJECTS

Real-Time Portfolio Rebalancing & Analytics Engine

- Built Python-based FastAPI microservices and Kafka/gRPC pipelines to automate rebalancing calculations, market-event processing, and advisor workflows with optimized PostgreSQL & Elasticsearch queries.
- Developed React + Redux dashboards for real-time analytics and integrated AWS Lambda, CloudWatch, and Step Functions to orchestrate serverless trading and data-reconciliation pipelines.
- Deployed end-to-end infrastructure using Docker, Kubernetes, and GitHub Actions CI/CD, with robust test coverage via PyTest/Jest and observability through Kibana and Redis caching.

Connected Automotive Diagnostics & Warranty Automation Platform

- Engineered Python Flask/FastAPI microservices with TypeScript/Node.js helpers for telematics processing, VIN decoding, and automated warranty validations, supported by Redis and Elasticsearch for low-latency retrieval.
- Implemented Kafka-streaming + gRPC services deployed on AWS EC2/EKS clusters, and built dealership-facing React interfaces for diagnostics, service history, and workflow automation.
- Achieved reliable continuous delivery using Docker, Kubernetes, GitLab CI/CD, and strengthened quality with Cypress testing, Jaeger tracing, and structured logging for production debugging.

CERTIFICATIONS

- AWS Certified Developer – Associate
- Oracle Certified – Database Design, Database Programming with PL/SQL, Database Programming with SQL